

Comms, GPS & Basestation

A Seaglider gets everything from its surface calls: position, data, and the next set of instructions. The chain is long: GPS and Iridium share the antenna mast in the rudder, a cable and bulkhead connector bring the signal into the hull, an RF relay switches it between GPS and modem, the Iridium modem makes the call, and a basestation on the other end has to accept the login and process the files. A weak link anywhere shows up the same way: calls that fail, and a glider that can't be told what to do.

Source

Paraphrased from 2017–2024 correspondence between Seaglider operators, APL-UW IOP, the manufacturer, satellite airtime providers and service providers (antenna failures, dial-up basestation debugging, the 2019 GPS rollover, Argos/backup trackers). Defer to APL-UW IOP for basestation software and firmware — see the [Electronics](#) page for the IOP links.

Call paths: RUDICS and dial-up

Gliders normally carry two numbers:

- **Primary (`telnum`)** — usually **RUDICS**. The call is routed from Iridium's gateway over the internet to a TCP port on the basestation. It is reliable, and the airtime rates are competitive.
- **Alternate (`altnum`)** — often a **dial-up (circuit-switched data)** call to a modem on a phone line, or a second RUDICS route.

Things to know when setting these up:

- **One RUDICS group delivers to one IP address.** For a second RUDICS number that goes to a different basestation, you need a second group, and that doubles the setup fee.
- **Dial-up over modern phone networks is fragile.** Iridium's analog data calls often fare badly across digital trunk lines, especially over long international routes. Some operators find it almost useless as a backup in their region.
- **An Iridium modem at the basestation** avoids the phone network. A modem with its own SIM, on a serial port of the basestation server, can take the